Robust flight navigation developed for complex vision based tasks

Massachusetts Institute of Technology's (MIT's) Computer Science and Artificial Intelligence Laboratory (CSAIL) researchers have developed robust flight navigation agents for vision based tasks in new and complex settings, inspired by organic brain adaptability. The researchers propose learning based control for drone adaptability across different environments without extra training. Liquid

Makram Chahine, a PhD student in electrical engineering and computer science and an MIT CSAIL affiliate, leads a drone used to test liquid neural networks (Credit: Mike Grimmett/MIT CSAIL)

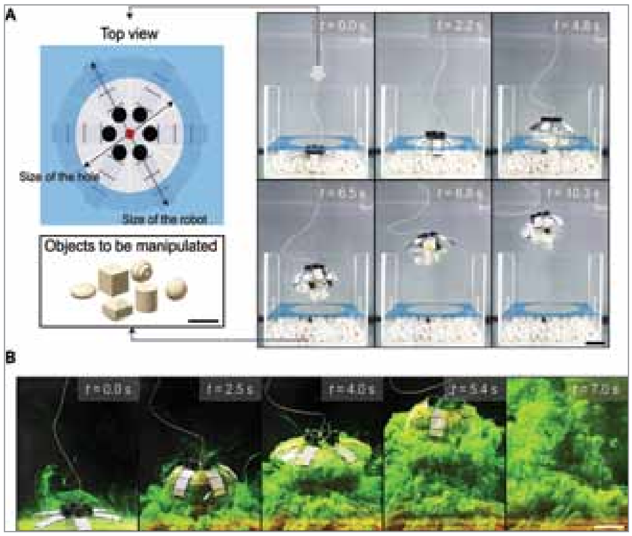
neural networks offer a promising solution to traditional deep learning's inability to capture causality, hindering adaptation to new environments. The team trained their system on pilot data and tested navigation skill transfer to new environments with changing conditions. Liquid neural nets' parameters can adapt over time, making them more robust to unexpected or noisy data. Liquid neural networks empower autonomous air mobility drones for environmental monitoring, package delivery, autonomous vehicles, and robotic assistants. Robust learning and performance in out-of-distribution tasks are crucial for machine learning and autonomous robots in critical societal applications.

Small wake-up receiver extends battery life from one day to one month

Massachusetts Institute of Technology (MIT) researchers have developed a smaller wake-up receiver using a few microwatts of power, including an authentication system that prevents battery-draining attacks. The team created a small chip using terahertz waves as a wake-up receiver for wireless communication with a far signal source. The researchers made a 1.54sq mm zero-power-consumption detector using dual antennas and a wake-up receiver. The chip, using less than 3 microwatts, converts terahertz waves to digital signals and activates the device with a matching token. Adding authentication blocks with randomised tokens to wake-up receivers can extend battery life from one day to one month and prevent hacker attacks. Researchers used lightweight cryptography and tested receiver sensitivity to terahertz signals up to 10 metres away. Terahertz beam-steerable arrays minimise signal loss and enable communication to multiple chips.

Underwater jellyfish-like robot cleans waste and collects fragile samples

Scientists at the Max Planck Institute for Intelligent Systems (MPI-IS) have developed a versatile, energy-efficient, and almost noiseless robot, reminiscent of a jellyfish, that can fit into the palm of a hand. Jellyfish-inspired robots collect samples quietly and safely, making them ideal for cleaning waste and collecting fragile samples without environmental harm. The robot has multiple layers, including



Contactless manipulation of irregular-shaped objects and dye. (A) Shape adaptation of the robot to transport objects of various shapes out of a confined space (f = 1Hz, D = 0.7; movie S2). The space was composed of a box and a plate on the top with a central hole. The diameter of the hole (13cm) was smaller than the outer diameter of the robot (16cm). (B) Visualised fluidic mixing of the fluorescein dye while the robot was swimming (f = 1Hz, D = 0.7; movie S1). Scale bars, 5cm (Credit: Science Advances, 2023)

stiffening and floating layers and a polymer skin, with HASELs embedded in the middle. HASELs are plastic pouches with liquid dielectric and charged electrodes. They contract and relax by pushing oil with electrical charges, sustaining high electrical stresses, and insulating against water. The team first developed Jellyfish-Bot with one electrode and six arms, then divided the electrode into groups to actuate them independently. This is the first underwater robot using HASELs.